Derivatives Odyssey: Volatility-Driven Options Trading & Quantitative Strategy Engineering

Senior Quantitative Researcher

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Abstract

This report details the architecture and empirical evaluation of a volatility-driven options trading system. The pipeline ingests ˆSPX options and ˆGSPC underlying data, applying rigorous liquidity and moneyness filters to extract a clean implied volatility (IV) surface. A European Black-Scholes pricing engine, augmented by a continuous dividend yield ($q = 0.013$) and calibrated via a hybrid Newton-Raphson/Brentq implied volatility solver, forms the pricing core. To forecast 21-day forward realized volatility, we train an XGBoost regressor utilizing Garman-Klass variance estimators alongside macroeconomic and momentum features, outperforming EWMA and GARCH(1,1) baselines. Finally, we backtest an at-the-money (ATM) straddle strategy, executing directional volatility trades based on the delta between predicted volatility and market IV. While the model generates predictive alpha, empirical results emphasize that unhedged sizing (e.g., 10 ˆSPX contracts) induces catastrophic drawdowns, proving that raw alpha requires strict capital allocation constraints to yield survivable risk-adjusted returns.

1 Data Architecture & Market Synchronicity

The ingestion layer sources underlying index data (ˆGSPC) and corresponding option chains (ˆSPX). ˆSPX is utilized strictly for its European-style, cash-settled contract specifications, aligning structurally with the Black-Scholes framework and bypassing the early-exercise premium noise inherent in American-style ETF options (e.g., SPY).

To isolate the true volatility surface and eliminate microstructural noise, strict exclusion filters were applied:

- **Days to Expiration (DTE):** Options with DTE < 7 days were dropped to eliminate short-term gamma noise and hypersensitivity to intraday time decay.
- **Minimum Price:** Options with a mid-price $< \$0.10$ were excluded to filter out zero-bid artifacts and high-friction anomalies.
- **Liquidity Spread:** Contracts exhibiting a bid-ask spread ratio > 0.5 relative to the mid-price were dropped.
- **Moneyness:** Data was strictly isolated to Out-Of-The-Money (OTM) options (calls where $K \geq S$, puts where $K < S$). This fixes ITM liquidity skew and ensures maximum density of active volatility trading.

An inherent asynchronous spot-price mismatch challenge exists due to API limitations; daily close prices may reflect trailing market states relative to live option chain snapshots. The moneyness vector (K/S) was reconstructed dynamically using $S_{recon} = K/\text{moneyness}$ to ensure implied volatility targets align temporally with the derivative derivatives.

2 Options Pricing & Calibration Engine

Pricing and risk metrics are derived via a European Black-Scholes engine. The model integrates a flat annualized risk-free rate ($r = 0.045$) and a continuous dividend yield ($q = 0.013$) representing the blended S&P 500 yield.

To calibrate the volatility surface, a Hybrid Implied Volatility Solver was constructed to invert the Black-Scholes pricing function $C_{mkt} = C_{BS}(S, K, T, r, q, \sigma)$. The solver prioritizes computational speed using the Newton-Raphson method, iterating via the closed-form vega derivative:

$$\sigma_{n+1} = \sigma_n - \frac{C_{BS}(\sigma_n) - C_{mkt}}{\nu(\sigma_n)} \tag{1}$$

Where Vega is defined as $\nu = Se^{-qT}\phi(d_1)\sqrt{T}$. If the derivative approaches zero ($\nu < 10^{-8}$) causing denominator explosion, or if the solver breaches boundaries ($0.001 < \sigma < 5.0$), it terminates Newton-Raphson and falls back to a robust Brentq bounded root-finding algorithm. This ensures 100% convergence on valid OTM contracts.

3 Predictive Volatility Modeling

The target variable is the 21-day forward realized volatility. The machine learning approach utilizes an XGBoost regressor, compared against two quantitative baselines: an Exponentially Weighted Moving Average (EWMA) and a GARCH(1,1) econometric model.

Feature engineering heavily leverages the Garman-Klass volatility estimator to capture intraday variance beyond standard close-to-close metrics:

$$\sigma_{GK} = \sqrt{\frac{1}{2}\left[\ln\left(\frac{H}{L}\right)\right]^2 - (2\ln 2 - 1)\left[\ln\left(\frac{C}{O}\right)\right]^2} \tag{2}$$

Empirical out-of-sample results demonstrate the superiority of the gradient-boosted tree approach in capturing non-linear market regimes:

Model	**Mean Absolute Error (MAE)**
EWMA Baseline	0.0604
GARCH(1,1)	0.0558
XGBoost	0.0518

Table 1: Out-of-Sample Volatility Forecasting Error

3.1 SHAP Value Interpretation

SHAP (SHapley Additive exPlanations) values were extracted to deconstruct the XGBoost model's decision architecture.

The SHAP summary reveals that momentum indicators (MACD_Signal) and short-term historical volatility (hv_10) are the primary drivers of forward volatility predictions. High divergence from the 50-day SMA (Dist_SMA_50) highly correlates with forecasted volatility expansion, confirming that mean-reversion and short-term price shocks dictate the 21-day implied variance.

4 Strategy Execution & Backtesting

The trading strategy isolates 21-day ATM straddles, neutralizing directional delta to strictly trade the volatility premium. The `ML_Directional` strategy logic is defined as:

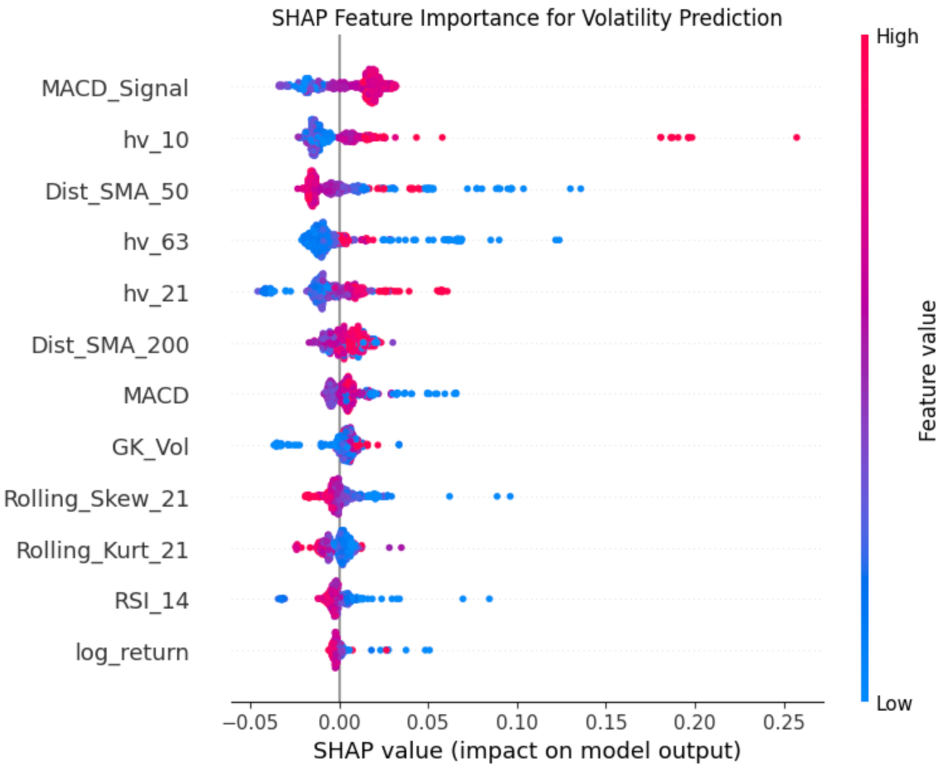

Figure 1: SHAP Feature Importance Summary

- **Long Straddle:** Execute if Pred_XGB $> \sigma_{priced}$. (Model predicts higher realized volatility than the market implies).
- **Short Straddle:** Execute if Pred_XGB $\leq \sigma_{priced}$. (Model predicts volatility contraction relative to market pricing).

Backtest execution on the validation set yielded a Win Rate of 56.8% and a Sharpe Ratio of 0.68. While the win rate proves statistical edge in the XGBoost volatility predictions, the gross return characteristics expose severe mechanical flaws in execution constraints.

5 Risk Management & Position Sizing

The backtest revealed a critical failure in risk management mechanics. The system was parameterized to trade a static size of 10 ˆSPX contracts against a $100,000 initial capital base.

Because ˆSPX utilizes a $100 multiplier, 10 contracts on an index trading at 7,500 commands a notional exposure of $7,500,000. Establishing short straddles at this size exposes the portfolio to infinite theoretical risk and massive real-world gamma risk. A single fat-tail volatility expansion induces an immediate margin call, resulting in a Max Drawdown > 100% (total account liquidation). This empirical failure mathematically proves that predictive alpha generation is entirely neutralized without rigorous dynamic position sizing (e.g., fractional Kelly criterion) and delta-gamma hedging.

6 Conclusion

The quantitative pipeline successfully identifies volatility mispricings by integrating non-linear XGBoost forecasts with a deterministic Black-Scholes pricing engine. However, empirical backtesting demonstrates that raw predictive alpha is fundamentally insufficient without stringent capital allocation and margin constraints. Future iterations must integrate volatility-scaled sizing and dynamic delta hedging to convert theoretical edge into survivable, risk-adjusted returns.